Nichols Group Executive Portal — Technical Architecture v0.3

Oct 18, 2026 · @Saurabh Mathur

Summary

The Nichols Executive Portal is a private, installable app (Mac and iPhone) that gives Robert one view across Nichols Law, Nichols Capital, Pub K Group, Ask Pub K, Pub K Institute and his personal accounts. All functionality is complete by December 31, 2026. The Financial Dashboard for all five firms, inbox, calendar, search and the daily briefing go live on December 22. The Approval Center, Commitments and Memory switch on December 29, after their own security test; every action that changes something in an outside system (sending an email, updating a task) requires Robert's approval.

The design rests on four rules:

- **Each firm is a separate lane.** Data, logins and AI context never mix between firms. The database itself enforces this.
- **Every number comes from the source systems.** QuickBooks, Stripe and TimeSolv figures are calculated by the database. Claude explains and drafts, but never produces figures.
- **Content is read live, not copied.** Email, chat and document bodies stay in Microsoft 365 and Google. The portal keeps only metadata and approved summaries.
- **Nothing happens without approval.** The app reads, analyzes and drafts freely, but any outside action needs Robert's explicit approval of the exact content.

Decisions requested from Robert

#	Decision	Recommendation
1	Approve this architecture and the stack: Vercel (hosting), Supabase (database), Inngest (background jobs), Claude API	Approve; all except Inngest are existing subscriptions
2	Hosting on commercial cloud instead of Azure, as assumed in the implementation memo	Approve, with Nichols Law isolated in its own database (Section 5)

#	Decision	Recommendation
3	Nichols Law fully in scope for 2026: finance plus read-only email, calendar, Teams and document search, with tenant details and admin consent from Integris	Approve, subject to Integris confirming the GCC setup (Section 5)
4	Full scope by Dec 31: finance, inbox, calendar, search and briefing live Dec 22; Approval Center, Commitments and Memory live Dec 29 after a security test	Approve
5	Production accounts (GitHub, Supabase, Inngest, Anthropic, Intuit, Google Cloud)	Approve

Scope

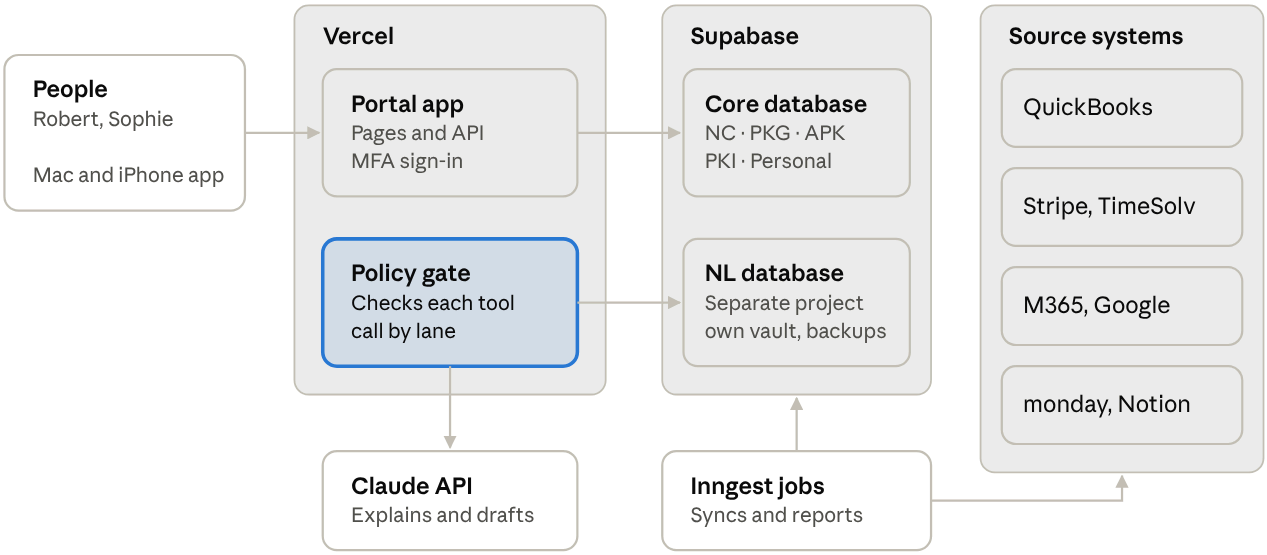
The portal has 13 modules. All 13 ship by December 31, 2026: ten go live on December 22, and the three action modules switch on December 29 after their own security test.

Module	What it does	Live
Platform core	Login with MFA, lanes, permissions, audit log, settings	Dec 22
Connector gateway	Secure access to M365 (including Nichols Law via Integris), Google, QuickBooks, Stripe, TimeSolv, monday.com, Notion	Dec 22
Business Performance (Finance)	Per-firm and consolidated dashboards, controls, weekly and monthly reports	Dec 22
Assistant	Ask Claude questions across permitted lanes, with sources	Dec 22
Communications	Priority inbox across Outlook, Gmail, Teams; reply drafts	Dec 22; sending via Approval Center Dec 29

Module	What it does	Live
Calendar & Meetings	Unified calendar, meeting prep packs, scheduling proposals	Dec 22; changes via Approval Center Dec 29
Documents & Search	Search across SharePoint, OneDrive, Drive, Notion; indexing of opted-in folders	Dec 22
Daily Briefing	Morning summary across lanes	Dec 22
Notifications	Push alerts on Mac and iPhone	Dec 22
Admin & Observability	Connection health, sync status, audit viewer	Dec 22
Approval Center	Approve, edit or reject every proposed action	Dec 29
Commitments	Track promises made in email, Teams and meetings	Dec 29
Memory & Preferences	Assistant remembers preferences per lane	Dec 29

Architecture at a glance

Firms stay separated; Claude reaches data only through the policy gate



system overview · app, policy gate, two databases, jobs, sources

People use one app; Claude reaches firm data only through the policy gate, which checks each request against the user's lanes. Background jobs pull data from the source systems into the Core and NL databases, so dashboards read from the database rather than calling vendors on every page load.

Technology stack

Only two new services are needed: Inngest for background jobs and a separate Supabase project for Nichols Law. Everything else uses subscriptions Nichols already has.

Layer	Choice	Why	Subscription
App	Next.js (TypeScript), installable web app (PWA)	One codebase for Mac and iPhone; no app store	—
Hosting	Vercel Pro	Preview per change, firewall, managed deployments	Existing

Layer	Choice	Why	Subscription
Database, login, file storage	Supabase: Core project + separate Nichols Law project	Row-level security enforces firm separation; MFA; encrypted token vault	Existing plan + added projects and backups
Background jobs	Inngest	Reliable scheduled syncs, retries, report runs	New
AI	Claude API (Anthropic), behind a provider interface	Analysis, drafting, classification; can be switched to Claude via Azure Foundry later	Existing
Secrets for people	1Password	Shared admin credentials, API keys handed over safely	Existing
UI	Tailwind, shadcn/ui, Recharts	Clean, consistent dashboards	Open source
Testing	Vitest, Playwright, pgTAP, prompt-injection suite	Separation and security proven by tests	Open source
Error tracking	Sentry	Alerts on failures	Free tier initially
Code and releases	GitHub (Nichols-owned), built in VS Code with Claude Code	Reviewed, reproducible releases	Existing

Firms, data separation and access

Each firm and Robert's personal accounts form a **lane** with its own logins, stored data, AI context and access rules. Nichols Law runs on a physically separate database; the other lanes share a Core database where row-level security blocks any cross-lane read.

Lane	Database	Sources in 2026	Notes
Nichols Law (NL)	Separate NL project	QuickBooks, TimeSolv, Billables (via TimeSolv), monday.com, M365 GCC tenant (via Integris)	Trust shown as reconciliation status, never client ledgers; email and documents read live, never stored
Nichols Capital (NC)	Core	QuickBooks (to confirm), M365 pilot	
Pub K Group (PKG)	Core	QuickBooks, Stripe, M365/Google pilot	
Ask Pub K (APK)	Core	Stripe; QuickBooks file or class (to confirm)	Can be its own QBO file or a class in Pub K's file
Pub K Institute (PKI)	Core	QuickBooks file or class (to confirm)	Same flexibility as Ask Pub K
Personal (P)	Core, owner only	Personal mail and calendar pilot	

Nichols Law and Integris. Integris manages the NL Microsoft 365 tenant, which runs on GCC. Nichols Law is fully in scope for 2026. Integris provides the tenant details (GCC or GCC High, conditional access, app-consent policy) and grants admin consent for the portal's Microsoft app. NL finance data comes from QuickBooks and TimeSolv through Sophie. NL email, Teams and documents are read live and never stored, and NL content is sent to Claude only after Robert confirms that the firm accepts Anthropic as a processor for client communications (zero data retention requested).

Who sees what.

Role	Access
Owner (Robert)	All lanes, consolidated view, all modules, settings
Finance operator (Sophie)	Finance module for all firms; generates and signs off reports
Viewer (Management Committee)	Named views only, e.g. NL weekly cash

Role	Access
Technical admin (Saurabh)	Connections, health, configuration; no financial or content access by default

The consolidated view shows summaries per firm only, never raw records, and Nichols Law appears there at status level.

Module design

Every module follows the same pattern: data synced or fetched through the connector gateway, rules enforced in the database, and AI tools that can read but never act on their own.

Module	Data and sources	How it works	Safeguards
Platform core	Users, lanes, memberships, audit log	Microsoft sign-in with enforced MFA; step-up MFA for sensitive actions	Append-only audit log; permissions tested automatically
Connector gateway	M365 (per tenant), Google, QuickBooks (per file), Stripe, TimeSolv, monday.com, Notion	One standard interface per source; tokens stored encrypted in each lane's vault	Tokens never reach the browser or Claude; one tenant's outage never blocks another
Business Performance	QuickBooks reports and transactions, Stripe, TimeSolv, monday.com finance boards	Scheduled syncs into Postgres; metrics calculated in SQL; weekly (Sat–Fri) and monthly snapshots locked after sign-off	Book balance never labeled as bank balance; no account numbers, compensation or trust client ledgers stored

Module	Data and sources	How it works	Safeguards
Assistant	Tools from each module	Claude answers using read-only tools; every claim cites its source and time	Server checks every tool call against the user's lanes; retrieved text cannot give instructions
Communications	Outlook, Gmail, Teams	Metadata index plus AI priority scoring; bodies fetched live when opened	Bodies not stored; NL included once Integris grants consent
Calendar & Meetings	M365 and Google calendars	Unified view; prep pack from recent threads, documents and open items	Changes go through the Approval Center
Documents & Search	SharePoint, OneDrive, Drive, Notion	Parallel live search, results merged with links	Respects source permissions; only opted-in folders indexed
Daily Briefing	All permitted modules	One summary per lane, then a combined briefing from those summaries	NL detail stays in its lane; the combined briefing gets a summary
Notifications	App events	Push to Mac and iPhone; in-app alerts	Push text carries no sensitive detail
Admin & Observability	Connections, sync runs, audit log	Health board, token expiry alerts, failed-job alerts	Technical admin sees no financial content
Commitments	Sent mail, meeting notes	AI extracts promises; matches them to monday.com/Notion tasks	Nothing created without approval

Module	Data and sources	How it works	Safeguards
Approval Center	Proposed actions	Exact content locked by hash; approve, edit (new proposal) or reject; executed once and verified	Payments, transfers, trust disbursements and deletions excluded
Memory	Per-lane preferences	Saved only from explicit statements; reviewable and deletable	Never shared across lanes

Data storage policy

The portal stores structured finance data and metadata; it does not store the contents of email, chats or documents.

Data	Stored?	Where	Retention
Login tokens and API keys	Yes, encrypted	Vault in each lane's database	Until revoked
Finance data (balances, invoices, bills, reports)	Yes	Lane database	Weekly and monthly snapshots 7 years
Email and chat bodies	No, read live	—	—
Email and chat metadata, priority scores	Yes	Lane database	90 days
Calendar event details	Yes (metadata)	Lane database	180 days
Document content	Only opted-in folders, as searchable excerpts	Lane database	Until source deleted or opt-out
Assistant conversations	Yes	Lane database	180 days, deletable by Robert
Approvals and audit log	Yes, cannot be altered	Lane database / Core	7 years

Data	Stored?	Where	Retention
Bank and routing numbers, SSNs, compensation tables, trust client ledgers	Never	—	—

Security and AI safeguards

Security is enforced by the database and the server, not by instructions to the AI, and every safeguard is covered by automated tests that run on each change.

Area	Control
Identity	Microsoft sign-in, MFA required for all data, short sessions, fresh MFA for approvals
Firm separation	Row-level security on every table; separate Nichols Law database; tests prove cross-lane reads fail
AI	Claude gets narrow, read-only tools; the server checks each tool call against the user's lanes; email and document text is treated as data and cannot trigger actions
Credentials	Source logins held in an encrypted vault per lane; never shown to users, logs or Claude
Actions	Proposal → Robert approves exact content → system re-checks → executes once → verifies → logs. Editing a proposal cancels the approval
Excluded actions	Payments, transfers, trust disbursements, QuickBooks posting, deletions and permission changes
Network	Vercel firewall and rate limits; signed webhooks only
Backups	Point-in-time recovery on production; restore drill before go-live

Area	Control
Testing	Internal security test before go-live: firm isolation, login and MFA, token handling, webhook spoofing, prompt injection
Ownership	All production accounts owned by a Nichols entity; the developer is a member, not the owner

Delivery, costs and risks

Release 1 goes live on December 22, 2026: design sign-off Nov 11, platform ready Nov 20, finance feature-complete Dec 4, figures tied out to the current manual reports for two weeks by Dec 18; the Approval Center, Commitments and Memory switch on Dec 29. The full task list with owners and dates is in the project plan:

Nichols Executive Portal — Project Plan to Dec 31, 2026

Running costs (new spend only).

Item	During build	In production
Supabase (added projects, backups)	~$35–50/month	~$250–300/month
Inngest	$0 (free tier)	~$50–150/month
Claude API usage	~$20–75/month	~$150–600/month
Sentry, Google Cloud	$0	~$0–35/month
Total	**~$55–125/month**	**~$450–1,100/month**

Vercel Pro, 1Password, the Claude API account and Microsoft 365 are existing subscriptions. Security testing is done internally.

Main risks.

Risk	Mitigation
Source data and process details take longer to assemble than planned	Templates issued at kickoff (Oct 21); build continues on synthetic data in parallel

Risk	Mitigation
Report and KPI definitions need more iteration	Start from the current SOP reports; refine after go-live
Intuit production approval takes longer than expected	Applied for in week 1; build on the QuickBooks sandbox
Figures need adjustment during tie-out	Three weeks of side-by-side checks before go-live
Nichols Law tenant (GCC) configuration needs extra steps	Raised at kickoff; NL inbox and calendar can follow finance without delaying it
Action modules need more security work	Separate security test Dec 17–23; actions switch on only when it passes
Hosting direction changes to Azure	Decision before any production resources exist; AI and hosting sit behind interfaces

Approval

Robert approves the architecture and the five decisions in the Summary; Saurabh confirms the technical design. Changes requested here are made before the database design is finalized.

Name	Role	Decision	Date	Comments
Robert Nichols	Sponsor	▾		
Saurabh Gopal	CTO / technical lead	▾		